



Customer Shopping Behaviour Analysis

3,900 transactions — product, demographic & behavioural insights
to inform product and marketing decisions

1. Project Overview

- Scope: 3,900 purchases across multiple categories
- Objectives: spending patterns, segments, product preference, subscription behaviour
- Outcome: actionable recommendations for subscription, pricing and targeting



2. Dataset Summary

Size & Shape

Rows: 3,900 · Columns: 18

Key Features

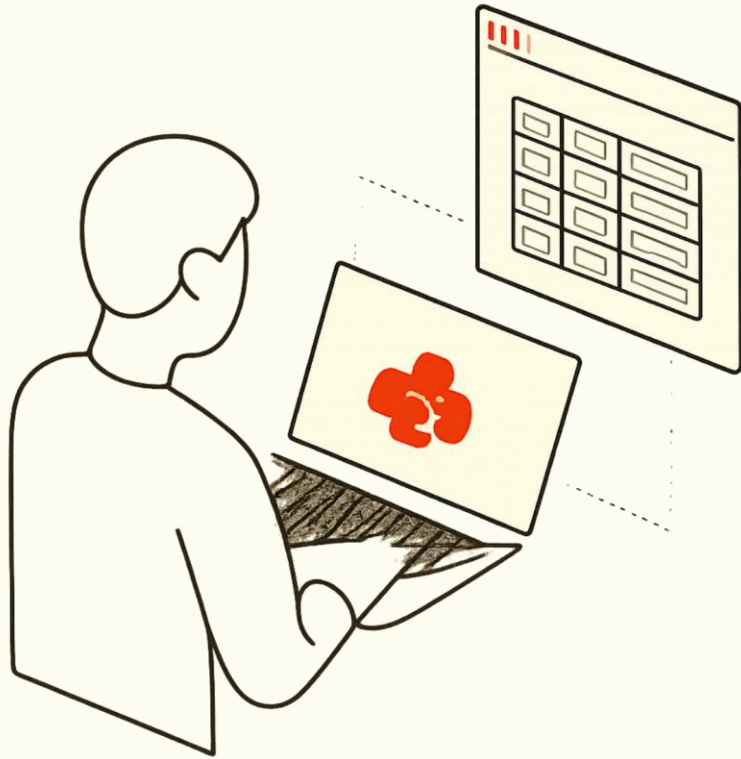
Demographics, Purchase details, Behavioural flags, Ratings, Shipping

Data Quality

37 missing values in Review Rating — imputed by category median



3. EDA — Python Workflow (summary)



O1

1. Load & Inspect

`df.info()`, `describe()` — structure + summary stats

O2

2. Clean & Impute

Median imputation for ratings; drop redundant `promo_code_used`

O3

3. Feature Engineering

`age_group` bins, `purchase_frequency_days`

O4

4. Persist

Export to PostgreSQL for SQL analytics

4. SQL Analysis — Key Findings (1/2)



Revenue by Gender

Male: \$157,890 · Female: \$75,191 — males account for ~68% of revenue



High-spend discount users

839 rows: customers using discounts but spending above average — target for loyalty upsell



Top Rated Items

Gloves, Sandals, Boots, Hat, Skirt — average ratings ~3.78–3.86

4. SQL Analysis — Key Findings (2/2)



Subscribers vs Non-subscribers

Subscribers: 1,053 customers · avg spend \$59.49 · revenue \$62,645

Non-subscribers: 2,847 customers · avg spend \$59.87 · revenue \$170,436

Discount-Dependent Products

Top: Hat (50%), Sneakers (49%), Coat (49%), Sweater (48%), Pants (47%)

5. Segmentation & Top Products



Loyal customers — largest cohort (purchase frequency high)



Returning customers



New customers

Top products by category (summary)

Accessories: Jewelry, Sunglasses, Belt

Clothing: Blouse, Pants, Shirt

Footwear: Sandals, Shoes, Sneakers

Outerwear: Jacket, Coat

📌 Use top items for promotional bundles and homepage highlights

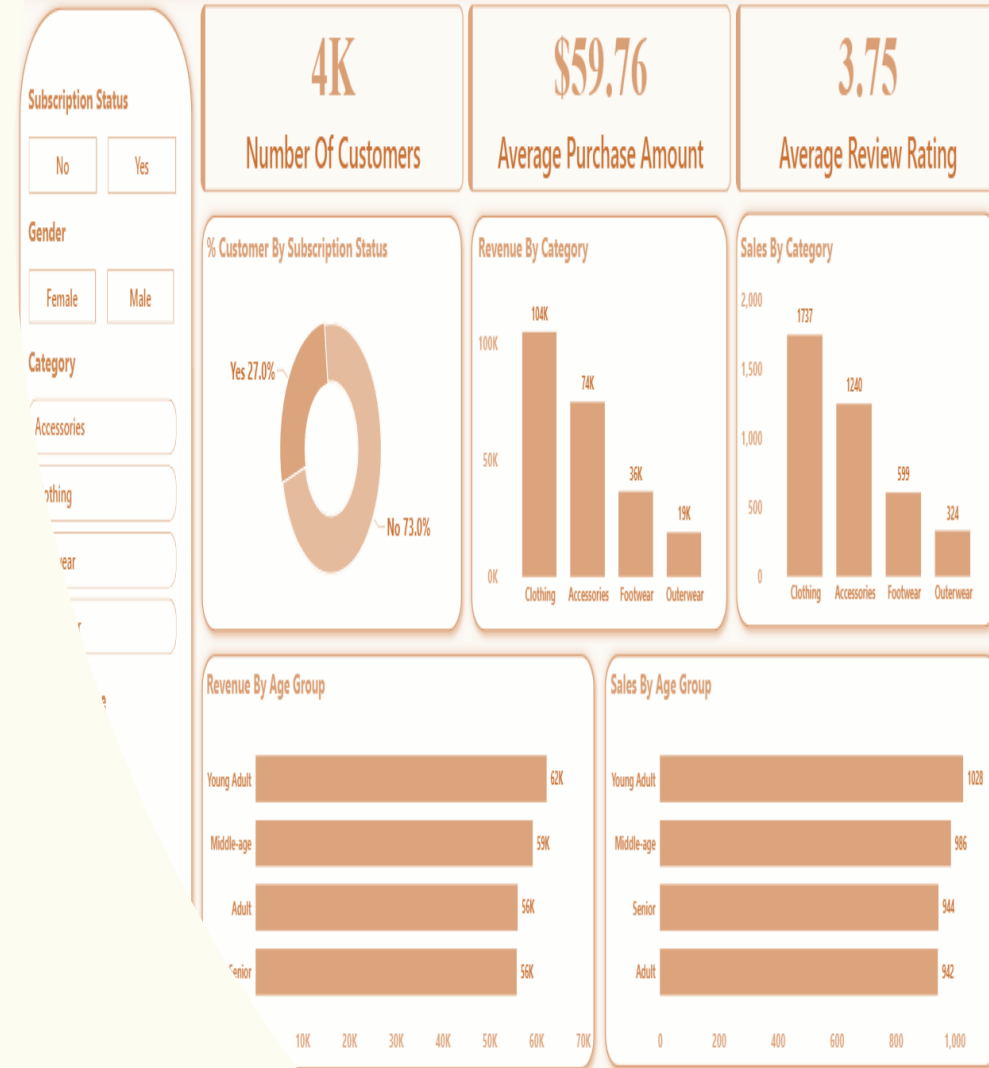
6. Dashboard — Power BI

Interactive dashboard built to explore:

- KPI cards: number of customers, avg purchase, avg rating
- Donut: subscription split (No 73% / Yes 27%)
- Bar charts: revenue & sales by category, by age group
- Filters: Gender, Category, Shipping, Subscription

Link: [Internal Power BI report \(embed for stakeholders\)](#)

Customer Behavior Dashboard



7. Business Recommendations

Boost Subscriptions

Exclusive benefits, targeted trial offers for high-frequency customers

Loyalty Program

Reward repeat buyers (tiers based on >5 purchases) to increase CLV

Discount Strategy

Limit discount-dependant SKUs; use targeted discounts for customer retention

Product Positioning

Promote top-rated & best-selling items in hero campaigns and bundles

Targeted Marketing

Prioritise Young Adults & Middle-age cohorts and express-shipping buyers

 Next steps: A/B test subscription incentives, refine discount thresholds, operationalise dashboard for weekly monitoring